

Operating System

□ Recognize the basic operating system concepts.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5132 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5132.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Engineering / Computer Hardware Engineering /
Course code	5132
Course title	Operating System
Semester	5 / 5 / 3 / 3 / 3 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

20 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ Recognize the basic operating system concepts.
- □ Provide a comprehensive introduction to process management, memory management, I/O management,

Course outcomes

CO	Official outcome	Study level
CO1	10 Understanding functions.	See official syllabus
CO2	Explain process and process management. 16 Understanding	See official syllabus
CO3	Illustrate memory management schemes. 20 Understanding Explain file organization and disk scheduling	See official syllabus
CO4	12 Understanding algorithms	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the significance of operating systems and its functions	4	As specified
Module 2	Describe process & process management	6	As specified
Module 3	Illustrate memory management schemes.	5	As specified
Module 4	Explain file organization and disk scheduling algorithms	5	As specified

MODULE 1

Outline the significance of operating systems and its functions

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the significance of operating systems and its functions

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

1 Understanding. application software. Explain System software-assembler, compiler,

1 Understanding. application software. Explain System software-assembler, compiler, is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. application software. Explain System software-assembler, compiler, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. application software. explain system software-assembler, compiler, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. application software. Explain System software-assembler, compiler, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഘട്ടം 1 Understanding. application software. Explain System software-assembler, compiler, ഡിഫിനിഷൻ/മിനിംഗ് ഡിഫിനിഷൻ/മിനിംഗ്. ഡിഫിനിഷൻ ഡിഫിനിഷൻ, steps, application ഡിഫിനിഷൻ ഡിഫിനിഷൻ. Technical terms English-ഘട്ടം ഡിഫിനിഷൻ ഡിഫിനിഷൻ.

interpreter and operating system with 3 Understanding. examples

interpreter and operating system with 3 Understanding. examples is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify interpreter and operating system with 3 Understanding. examples using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpreter and operating system with 3 understanding. examples should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What interpreter and operating system with 3 Understanding. examples means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-ഇന്റർപ്രീറ്റർ and operating system with 3 Understanding. examples ഉദാഹരണങ്ങൾ definition/meaning നിർവ്വചന/അർത്ഥം. ഉപയോഗം ഉപയോഗം ഉപയോഗം, steps, application ഉപയോഗം ഉപയോഗം ഉപയോഗം. Technical terms English-ഇന്റർപ്രീറ്റർ ഉപയോഗം.

Explain the functions of Operating System 3 Understanding.

Explain the functions of Operating System 3 Understanding. is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the functions of Operating System 3 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the functions of operating system 3 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the functions of Operating System 3 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- Explain the functions of Operating System 3 Understanding. definition/meaning . steps, application . Technical terms English- .

Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.

Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise different types of operating system 3 understanding. introduction to software: system software & application software, examples-system software - assembler, loader, compilers, interpreters and their functions, comparison. operating system - definition - functions of operating system - different types of

operating systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software – assembler, loader, compilers, interpreters and their functions, comparison. Operating system – definition – functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software – assembler, loader, compilers, interpreters and their functions, comparison. Operating system – definition – functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Outline the significance of operating systems and its functions
Explain software, system software and

M1.01 Understanding. 1

application software.

M1.02 Explain System software-assembler, compiler, interpreter and operating system with 3
Understanding. examples

M1.03 Explain the functions of Operating System 3
Understanding.

M1.04 Summarise different types of operating system 3
Understanding.

Contents:

Introduction to software: System software & application software, examples-System software – assembler, loader, compilers, interpreters and their functions, comparison.

Operating system – definition – functions of Operating system - different types of Operating

Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Describe process & process management

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe process & process management
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Understanding structure Explain various states of a process with state

2 Understanding structure Explain various states of a process with state is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding structure Explain various states of a process with state using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding structure explain various states of a process with state should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding structure Explain various states of a process with state means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- 2 Understanding structure Explain various states of a process with state definition/meaning. states, application. steps, application. Technical terms English-.

2 Understanding diagram Classify CPU schedulers and Illustrate

2 Understanding diagram Classify CPU schedulers and Illustrate is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding diagram Classify CPU schedulers and Illustrate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding diagram classify cpu schedulers and illustrate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding diagram Classify CPU schedulers and Illustrate means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- 2 Understanding diagram Classify CPU schedulers and Illustrate definition/meaning .
 , steps, application . Technical terms English- .

6 Understanding scheduling algorithms

6 Understanding scheduling algorithms is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding scheduling algorithms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding scheduling algorithms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding scheduling algorithms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 6 Understanding scheduling algorithms

എസ്കൂളിംഗ് അൾഗരിതം definition/meaning എസ്കൂളിംഗ് അൾഗരിതം. എസ്കൂളിംഗ് അൾഗരിതം എസ്കൂളിംഗ്, steps, application എസ്കൂളിംഗ് അൾഗരിതം എസ്കൂളിംഗ്. Technical terms English- എസ്കൂളിംഗ് എസ്കൂളിംഗ്.

Show resource allocation graph

Show resource allocation graph is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Show resource allocation graph using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, show resource allocation graph should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Show resource allocation graph means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Show resource allocation graph
 definition/meaning . . . , steps, application Technical terms English-
 .

Demonstrate deadlock and its causes

Demonstrate deadlock and its causes is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate deadlock and its causes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate deadlock and its causes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate deadlock and its causes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-☐☐ Demonstrate deadlock and its causes

ৱিকিপিডিয়াৰ পৰা definition/meaning গ্ৰহণ কৰা হৈছে. ৱিকিপিডিয়াৰ পৰা steps, application গ্ৰহণ কৰা হৈছে. Technical terms English-ৰ পৰা গ্ৰহণ কৰা হৈছে.

Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes-Process synchronization and critical section management.

Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes- Process synchronization and critical section management. is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes-Process synchronization and critical section management. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain process synchronization 2 understanding define process - process control block (pcb) and its general structure - different states of a process with the help of state diagram.- schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- various scheduling criteria - fcfs, sjf, priority, and rr scheduling algorithms with gantt charts -comparison of various scheduling algorithms-resource allocation graph-deadlock-its causes-process synchronization and critical section management. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes-Process synchronization and critical section management. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes-Process synchronization and critical section management. definition/meaning . steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Describe process & process management		
	Illustrate process control block with its	
M2.01		2
Understanding	structure	
	Explain various states of a process with state	
M2.02		2
Understanding	diagram	
	Classify CPU schedulers and Illustrate	
M2.03		6
Understanding	scheduling algorithms	
M2.04	Show resource allocation graph	2
Understanding		
M2.05	Demonstrate deadlock and its causes	2
Understanding		
M2.06	Explain Process synchronization	2
Understanding		
	Series Test – I	1
Contents:		
Define process - process control block (PCB) and its general structure -		
different states of a		
process with the help of state diagram.- Schedulers – long, medium and short		
term-		
difference between preemptive and non-preemptive scheduling- Various scheduling		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Illustrate memory management schemes.

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate memory management schemes.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Define memory management.

Define memory management. is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define memory management. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define memory management. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define memory management. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Define memory management. Explain the definition/meaning of memory management. Explain the steps, application of memory management. Technical terms English- Malayalam.

Show various address binding schemes

Show various address binding schemes is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Show various address binding schemes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, show various address binding schemes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Show various address binding schemes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Show various address binding schemes
പ്രതിബദ്ധനാശനം നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക,
steps, application നിർവ്വചനം നിർവ്വചിക്കുക. Technical terms English- നിർവ്വചനം
നിർവ്വചിക്കുക.

Explain memory management techniques

Explain memory management techniques is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain memory management techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain memory management techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain memory management techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain memory management techniques
memory management techniques definition/meaning memory management techniques, steps, application memory management techniques. Technical terms English- memory management techniques.

Illustrate memory allocation strategies.

Illustrate memory allocation strategies. is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate memory allocation strategies. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate memory allocation strategies. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate memory allocation strategies. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓരോന്നും Illustrate memory allocation strategies.

ഓരോന്നും definition/meaning ഓരോന്നും. ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നും. Technical terms English-ഓരോന്നും ഓരോന്നും.

Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings - compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation - fixed partition and variable partition - Allocation Strategies - first fit, best fit and worst fit - Define fragmentation - internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing.

Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings - compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation - fixed partition and variable partition - Allocation Strategies - first fit, best fit and worst fit - Define fragmentation - internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing. is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings - compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation - fixed partition and variable partition - Allocation Strategies - first fit, best fit and worst fit - Define fragmentation - internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate page replacement algorithms. 6 understanding memory management - different address bindings - compile, link and run time bindings. - difference between logical address and physical address - contiguous memory allocation - fixed partition and variable partition - allocation strategies - first fit, best fit and worst fit - define fragmentation - internal and external, and solutions - paging and paging hardware - segmentation, advantages of segmentation over paging- concept of virtual memory - demand paging - page-faults and how to handle page faults. - page replacement algorithms: fifo, optimal, lru -thrashing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings - compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation - fixed partition and variable partition - Allocation Strategies - first fit, best fit and worst fit - Define fragmentation - internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings - compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation - fixed partition and variable partition - Allocation Strategies - first fit, best fit and worst fit - Define fragmentation - internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal,

LRU -Thrashing. ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ definition/meaning ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ. ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ, steps, application ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ. Technical terms English-ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ ലാസ്റ്റ് റീയൂസ്ഡ് ഫസ്റ്റ് ഇൻ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Illustrate memory management schemes.

M3.01	Define memory management.	1
Remembering		
M3.02	Show various address binding schemes	3
Understanding		
M3.03	Explain memory management techniques	5
Understanding		
M3.04	Illustrate memory allocation strategies.	5
Understanding		
M3.05	Demonstrate page replacement algorithms.	6
Understanding		

Contents:

Memory management - Different address bindings – compile, link and run time bindings. -

Difference between logical address and physical address - Contiguous memory allocation –

fixed partition and variable partition – Allocation Strategies - first fit, best fit and worst fit -

Define fragmentation – internal and external, and solutions - Paging and paging hardware -

Segmentation, advantages of segmentation over paging- Concept of virtual memory -

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Explain file organization and disk scheduling algorithms

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain file organization and disk scheduling algorithms
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

2 Remembering operations

2 Remembering operations is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering operations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering operations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering operations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-2 Remembering operations $\square \square \square \square \square \square \square \square \square$
 $\square \square \square \square$ definition/meaning $\square \square \square \square \square \square \square \square \square \square$. $\square \square \square \square \square$ $\square \square \square \square \square$ $\square \square \square \square \square \square$, steps,
application $\square \square \square \square \square$ $\square \square \square \square \square \square \square \square \square \square$. Technical terms English- $\square$ $\square \square \square \square \square$
 $\square \square \square \square \square \square \square \square \square$.

Explain different file organizations

Explain different file organizations is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different file organizations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different file organizations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different file organizations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain different file organizations

Explain different file organizations. Define/meaning of file organization. Steps, application of file organization. Technical terms English-Malayalam.

Classify directory structures

Classify directory structures is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify directory structures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify directory structures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify directory structures means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Illustrate file allocation methods

Illustrate file allocation methods is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate file allocation methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate file allocation methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate file allocation methods means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓരോന്നും Illustrate file allocation methods ഓരോന്നും
definition/meaning ഓരോന്നും. ഓരോന്നും ഓരോന്നും steps,
application ഓരോന്നും ഓരോന്നും. Technical terms English-ഓരോന്നും
ഓരോന്നും.

Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm

Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain disk scheduling algorithms. 3 Understanding File system
 - Concept of file and directory - Various file operations - File organization concepts
 - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked

and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain disk scheduling algorithms. 3 understanding file system - concept of file and directory - various file operations - file organization concepts - sequential and indexed. different directory structures - single level, two-level, and tree structured directories. - different allocation methods - contiguous, linked and indexed allocations. various disk scheduling algorithms-fcfs, sstf, scan, c-scan, look & c-look. text / reference t/r book title/author abraham silberschatz, peter gaer galvin and greg gagne, operating system t1 concepts, wiley publications-eighth edition william stallings, operating systems-internals and design principles, r1 pearson publications- seventh edition r2 andrew s tanenbaum, herbert bos, modern operating systems, pearson, 4/e, online resources sl.no website link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105214/# 2 https://www.tutorialspoint.com/operating_system/index.htm means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain disk scheduling algorithms. 3

Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1

<https://nptel.ac.in/courses/106/105/106105214/#> 2

https://www.tutorialspoint.com/operating_system/index.htm definition/meaning steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Explain file organization and disk scheduling algorithms
Define the concept of file system & file

M4.01		2
Remembering		
	operations	
M4.02	Explain different file organizations	2
Understanding		
M4.03	Classify directory structures	2
Understanding		
M4.04	Illustrate file allocation methods	3
Understanding		
M4.05	Explain disk scheduling algorithms.	3
Understanding		
	Series Test – II	1

Contents:

File system - Concept of file and directory - Various file operations - File organization concepts – sequential and indexed. Different directory structures – single level, two-level, and tree structured directories. - Different allocation methods – contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look.

Text / Reference

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding. application software. Explain System software-assembler, compiler,. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. application software. Explain System software-assembler, compiler,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain interpreter and operating system with 3 Understanding. examples. (3 marks)

Answer: Begin with the standard definition or identity of *interpreter and operating system with 3 Understanding. examples.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the functions of Operating System 3 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the functions of Operating System 3 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions*

of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding structure Explain various states of a process with state. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding structure Explain various states of a process with state*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding diagram Classify CPU schedulers and Illustrate. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding diagram Classify CPU schedulers and Illustrate*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 6 Understanding scheduling algorithms. (3 marks)

Answer: Begin with the standard definition or identity of 6 *Understanding scheduling algorithms*. State the governing principle, list the key components or stages, and close

with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Show resource allocation graph. (3 marks)

Answer: Begin with the standard definition or identity of *Show resource allocation graph*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Define memory management.. (3 marks)

Answer: Begin with the standard definition or identity of *Define memory management..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Show various address binding schemes. (3 marks)

Answer: Begin with the standard definition or identity of *Show various address binding schemes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

application application application application.

Define or explain Explain memory management techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Explain memory management techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: application definition, principle/steps, application application.

Define or explain Illustrate memory allocation strategies.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate memory allocation strategies..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: application definition, principle/steps, application application.

Module 4

Define or explain 2 Remembering operations. (3 marks)

Answer: Begin with the standard definition or identity of *2 Remembering operations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: application definition, principle/steps, application application.

Define or explain Explain different file organizations. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different file organizations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Classify directory structures. (3 marks)

Answer: Begin with the standard definition or identity of *Classify directory structures*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Illustrate file allocation methods. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate file allocation methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding. application software. Explain System software-assembler, compiler,**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding structure Explain various states of a process with state.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Define memory management..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Remembering operations.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/106/105/106105214/#>
- https://www.tutorialspoint.com/operating_system/index.htm

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5132. Verify institutional updates against the current official curriculum.